Alfie R. Pelicano

Full-Stack Developer

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Professional Summary

Full-stack developer with 15+ years in scientific computing, now building production-grade web applications with modern JavaScript frameworks. Transitioned from GNSS data analysis and geodetic automation to delivering tested, accessible, and secure web platforms.

Currently maintaining 7 production applications across B2B marketplaces (PipetGo), queue management systems (Washboard), and community platforms (carpool coordination, parking management). Technical focus: Next.js, React, TypeScript, PostgreSQL, with emphasis on test-driven development (233+ tests across projects), WCAG 2.1 AA accessibility, and database-level security (RLS policies, transaction isolation).

My background in scientific workflows informs architectural decisions: I build for maintainability, data integrity, and long-term operational reliability. Every project ships with comprehensive test coverage, security-first authentication, and production-ready deployment pipelines.

Core competencies: Full-stack web development, test-driven development, session-based authentication, database design (PostgreSQL with Prisma), API development, accessibility compliance, and deployment automation (Vercel, Neon, Render).

Technical Skills

Frontend	Next.js 14 (App Router), React 18/19 (Server Components), TypeScript (strict mode), Vanilla JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, shadcn/ui, Responsive Design, WCAG 2.1 AA Accessibility
Backend	Node.js, Express, Next.js API Routes, PostgreSQL (advanced SQL), Prisma ORM, Database Migrations, Session-based Authentication, NextAuth.js (v4 & v5), RESTful API Design, Zod Validation
Database	PostgreSQL, NeonDB (serverless), Supabase, Row Level Security (RLS), Transaction Isolation (SERIALIZABLE), Database Triggers, Query Optimization, Connection Pooling, Idempotent Migrations
Testing & QA	Test-Driven Development (TDD), Vitest (unit testing), Playwright (E2E), Jest, React Testing Library, pg-mem (mock database), 90%+ test coverage standards, Security auditing, Manual QA process design
Python & Scientific	Python 3.11+, Click CLI, Rich Terminal UI, Pandas, Pytest, Shell Scripting (Bash), Fortran (10+ years - university and GPS geodesy post-processing), GNSS Data Processing, Ollama (local LLM orchestration), Scientific automation pipelines
DevOps	Git/GitHub workflows, Vercel deployment, Render, GitHub Pages, Environment configuration, CI/CD principles, Database migration strategies, Production monitoring
Security	Session management (httpOnly cookies, bcrypt), Rate limiting, SQL injection prevention (parameterized queries), Timing-safe comparisons, CSRF mitigation, Password hashing (bcrypt cost factor 12), Magic link authentication, OWASP Top 10

Architecture	Layered architecture (API → Service → Database), Multi-role authorization, Multi-tenant data isolation, Database-level constraints (EXCLUDE, CHECK), Error handling patterns, Documentation-driven development
Domain Expertise	GNSS/GPS data analysis, Geodetic automation, RINEX file formats, Scientific data workflows, Geophysics, Tectonic monitoring, Linux system administration, Bernese GNSS Software
Soft Skills	Analytical thinking, Problem-solving, Data interpretation, Technical documentation, Quality focus (QA mindset), Test case development, Bug reproduction, Team collaboration, Bilingual: English & Filipino

Production Applications

2024–Present **PipetGo – B2B Laboratory Testing Marketplace**, *Next.js 14.2 · React 18 · TypeScript · PostgreSQL (Neon) · Prisma · NextAuth · Vitest · Playwright*

Production RFQ platform connecting businesses with ISO 17025 certified labs across the Philippines. Implements multi-role authorization (CLIENT, LAB_ADMIN, ADMIN) with quotation workflow from request to fulfillment.

- 233 passing tests (Vitest + Playwright E2E)
- Request-for-Quote system with three pricing modes (QUOTE_REQUIRED, FIXED, HYBRID)
- File upload system (UploadThing) for specifications and test results
- Zod validation schemas for all API endpoints
- Session-based authentication with NextAuth 4.24.7
- Dual-mode testing (mock DB with pg-mem + live PostgreSQL)

Impact: Centralized directory of accredited labs replacing scattered, outdated data sources.
Repository: https://github.com/alfieprojectsdev/pipetgo

2024–Present **Washboard – Car Wash Queue Management System**, *Next.js 14 · TypeScript · PostgreSQL (NeonDB) · Session Auth · Vitest · pg-mem*

Production deployment at washboard.ithinkandicode.space replacing manual pen-and-paper queue tracking with contactless QR bookings and real-time dashboard.

- 130/130 tests passing with 3-5 second execution time
- 94/100 security audit score
- WCAG 2.1 AA accessibility compliant
- Layered architecture: API → Service → Database
- Transaction-safe queue management (SERIALIZABLE isolation level)
- Magic link system with 128-character cryptographically secure tokens
- Timing-safe token comparison (security best practice)

Impact: Piloting with car wash operators; eliminated manual position tracking and reduced peak-hour wait confusion.
Live Demo: https://washboard.ithinkandicode.space

2024 **Parkboard – Peer-to-Peer Parking Marketplace**, *Next.js 14 · TypeScript · PostgreSQL (Neon) · NextAuth v5 · Tailwind + shadcn/ui · Jest · Playwright*

Production-ready MVP for community parking slot sharing with flexible scheduling and real-time availability.

- 85% test coverage (158 unit tests, 8 E2E scenarios)
- Database-level overlap prevention (PostgreSQL EXCLUDE constraints)
- Server-side price calculation via database triggers (never trust client)
- Row Level Security (RLS) policies for multi-tenant data isolation
- Idempotent database migrations for safe schema evolution
- NextAuth.js v5 with email/password + OAuth support

Impact: Deployed for LMR residential community, replacing disorganized chat-based coordination.
Deployment: parkboard.app/LMR (community access)

2024 **Carpool Coordinator**, *Node.js · Express · PostgreSQL · Vanilla JavaScript · HTML5 · CSS3*

Production deployment on Render for Phirst Park Homes community carpooling.

- Structured posting system with route matching (Dau, Cubao, BGC, Manila)
- Contact integration (Messenger, Viber, Phone, Telegram)
- RESTful API with PostgreSQL backend
- Responsive design for mobile-first user base

Impact: Improved participation rates and reduced coordination time for parents and commuters.
Repository: https://github.com/alfieprojectsdev/carpool-app

2023–Present **Window Cards Generator – Math Worksheet Tool**, *Vanilla JavaScript · HTML5 · CSS3 · Print CSS*

Production tool for SPED and primary educators generating customizable arithmetic worksheets.

- Client-side only (no server dependency)
- Auto-generated answer keys
- Configurable operand lengths (4-6 digits), all four operations
- Iteratively refined based on real teacher feedback

Impact: Used in classrooms as free alternative to commercial workbook sets.
Live Demo: https://alfieprojectsdev.github.io/windowcards

Active Development Projects

2024–Present **Drive Archaeologist – GNSS Data Organization CLI**, *Python 3.11+ · Click CLI · Rich Terminal · Pandas · Pytest*

Phase 1/4 implementation for excavating and organizing decades of legacy GNSS data from old drives with inconsistent structures.

- RINEX file type detection with regex pattern matching
- MD5-based duplicate detection across drives
- Resume capability for long-running scans (progress tracking)
- Non-destructive by design (dry-run first, generate migration scripts)
- Designed for PHIVOLCS (Philippine Institute of Volcanology and Seismology)

Target Users: Geodesists, geophysicists managing legacy scientific archives
Repository: https://github.com/alfieprojectsdev/drive-archaeologist

2024–Present **NightCoder Orchestrator – ADR-to-Patch Automation**, *Python 3.11+ · Ollama · deepseek-r1 (14B) · qwen2.5-coder (14B) · Shell Scripting*

Phase 3/4 production-ready multi-agent automation engine that reads Architecture Decision Records (ADRs), generates implementation plans, and produces git-style patches using local LLMs.

- Multi-agent workflow: ADR parsing → Plan synthesis → Patch generation
- Local LLM orchestration via Ollama (no API costs)
- 95% plan quality, 90% patch cleanliness (validated)
- Safety-gated workflow with error handling
- Portable configuration for cross-repo use

Impact: Recent optimizations achieved production-ready reliability for automated patch generation.

2024–Present **SciWriter Orchestrator – Research Writing Pipeline**, *Python 3.11+ · Ollama · qwen2.5 (3B Instruct) · LaTeX · Beamer*

Phase 2/4 automated writing pipeline converting research notes into structured drafts with refinement passes, terminology checks, and LaTeX export.

- Multi-stage workflow: draft → refine → terminology normalization → export
- LaTeX and Beamer slide generation
- Terminology database (JSON) for consistency
- Config-driven pipeline for customization

Status: Pipeline functional, improving output consistency for academic paper generation.

Work Experience

2009–Present **Science Research Specialist**, *Philippine Institute of Volcanology and Seismology (PHIVOLCS)*

- Processed GNSS data for tectonic monitoring and earthquake hazard assessment across the Philippines
- Wrote and maintained Bash/Python scripts to automate geodetic campaign workflows and streamline repetitive data tasks
- Participated in GNSS field surveys and network maintenance, supporting station integrity and data continuity
- Contributed as co-author to peer-reviewed publications on crustal deformation and geodetic methods, with work cited in regional tectonics research
- Supported internal digitization and workflow modernization efforts through scripting, documentation, and hands-on technical support
- Transitioned technical skills to modern web development, building production applications that apply scientific rigor (TDD, data integrity) to software engineering (2022–Present)

2005–2009 **Software Test Engineer**, *Azeus Systems Philippines*

- Performed functional and regression testing for large-scale web applications, including government contract platforms
- Wrote detailed manual test cases and reported issues with clear reproduction steps, improving dev handoff quality
- Used SQL for test setup and validation of back-end logic and edge cases during integration testing
- Worked closely with developers and other testers in a structured QA pipeline

2004–2005 **Science Information Analyst**, *ESS Manufacturing Company, Inc. (EMCI)*

- Indexed and summarized physics and engineering articles for an international scientific publication database
- Applied technical reading and abstract writing skills under tight deadlines across multiple disciplines

2002–2003 **Student Assistant**, *National Institute of Physics – UP Diliman (Theoretical Physics Group)*

- Maintained a small number of Linux workstations for basic network access, file management, and local simulations
- Helped students troubleshoot basic command-line issues, particularly Fortran compilation and plotting with gnuplot
- Assisted with recovering and digitizing legacy scientific code from non-machine-readable sources

Education

1999–2004 **BS Applied Physics, Instrumentation**, *University of the Philippines Diliman*
Eugenio Lopez Jr. Scholar (2002–2004)

1995–1999 **Electronics Technology Track**, *Don Bosco Technical College*

2015 **Master of Geoscience (Partially Completed)**, *Macquarie University*
Completed a semester-length unit on scientific data scripting and visualization, covering UNIX shell, GMT, Terragen, Blender, and Python-based workflows

Select Publications & Research

2023 *Geophysical Research Letters:* Oblique blind faulting, Abra Earthquake

2013 *Journal of Geophysical Research: Solid Earth:* Interseismic deformation along the Philippine Fault

2015 *Journal of Disaster Research:* Plate convergence and GPS motions in Mindanao

2010–2013 *AGU Fall Meeting Abstracts:* Deformation modeling and GPS-based intrusion studies

(Full list available on request or via LinkedIn/Google Scholar profile)